Column 3 has no pivots. ∴ By defn, x_3 is a free variable.
By Theorem 2, the system of eqns. has infinitely many solutions.
∴ A nontrivial soln. to the matrix equation $Ax=0$ exists.

~~When h=6, the system of equati~~

So, when $h=6$, v_1, v_2 and v_3 are linearly dependent

We can similarly prove that when $h \neq 6$, v_1, v_2 and v_3 are linearly independent.